SNES CPU-visible register map

Every register the 65816 can reach directly (`$2100–$21FF` , `$4016–$43FF`), generated from the annotated HAL headers in `platforms/snes/snes_*.h` — the single source of truth. The audio processor (SPC700 + S-DSP) is reachable only through the four APU ports below and its internal registers are a separate address space (out of scope here; see the hardware summary). For the CPU core itself see the 65816 reference.

Bit-field facts are per nocash *fullsnes*, the SNESdev wiki, and anomie's docs (SOURCES.md).

PPU — Picture Processing Unit ($2100–$213F)

Display, backgrounds, OAM, VRAM, Mode 7, CGRAM, windows, colour math, and the read ports. — `snes_ppu.h` , 66 registers.

Addr	Name	R/W	Purpose
`$2100`	`REG_INIDISP`	W	Display control 1 — force-blank + master brightness.
`$2101`	`REG_OBSEL`	W	Object (sprite) size + tile-base select.
`$2102`	`REG_OAMADDL`	W	OAM word address, low byte (0..255).
`$2103`	`REG_OAMADDH`	W	OAM address high + priority rotation.
`$2104`	`REG_OAMDATA`	W	OAM data write (auto-increments OAMADD; write low then high).
`$2105`	`REG_BGMODE`	W	BG mode + per-layer tile size.
`$2106`	`REG_MOSAIC`	W	Mosaic block size + per-BG enable.
`$2107`	`REG_BG1SC`	W	BG1 tilemap base + map size.
`$2108`	`REG_BG2SC`	W	BG2 tilemap base + size (see BG1SC).
`$2109`	`REG_BG3SC`	W	BG3 tilemap base + size (see BG1SC).
`$210A`	`REG_BG4SC`	W	BG4 tilemap base + size (see BG1SC).
`$210B`	`REG_BG12NBA`	W	BG1/BG2 character (tile) data base.

Addr	Name	R/W	Purpose
`$210C`	`REG_BG34NBA`	W	BG3/BG4 character data base.
`$210D`	`REG_BG1HOFS`	W	BG1 horizontal scroll (write twice; 10-bit, +Mode7 share).
`$210E`	`REG_BG1VOFS`	W	BG1 vertical scroll (write twice; 10-bit).
`$210F`	`REG_BG2HOFS`	W	BG2 horizontal scroll (write twice).
`$2110`	`REG_BG2VOFS`	W	BG2 vertical scroll (write twice).
`$2111`	`REG_BG3HOFS`	W	BG3 horizontal scroll (write twice).
`$2112`	`REG_BG3VOFS`	W	BG3 vertical scroll (write twice).
`$2113`	`REG_BG4HOFS`	W	BG4 horizontal scroll (write twice).
`$2114`	`REG_BG4VOFS`	W	BG4 vertical scroll (write twice).
`$2115`	`REG_VMAIN`	W	VRAM address increment control.
`$2116`	`REG_VMADDL`	W	VRAM word address, low byte.
`$2117`	`REG_VMADDH`	W	VRAM word address, high byte.
`$2118`	`REG_VMDATAL`	W	VRAM data, low byte (increments per VMAIN if INCHIGH=0).
`$2119`	`REG_VMDATAH`	W	VRAM data, high byte (increments per VMAIN if INCHIGH=1).
`$2116`	`REG_VMADD` [16]	W	VRAM word address, 16-bit access alias over VMADDL/H.
`$2118`	`REG_VMDATA` [16]	W	VRAM data, 16-bit access alias over VMDATAL/H (low then high).
`$211A`	`REG_M7SEL`	W	Mode 7 screen settings.
`$211B`	`REG_M7A`	W	Mode 7 matrix A (signed 8.8; write low,high). Also hw-multiply factor A.
`$211C`	`REG_M7B`	W	Mode 7 matrix B (signed 8.8; write low,high). Also hw-multiply factor B (8-bit).

Addr	Name	R/W	Purpose
`$211D`	`REG_M7C`	W	Mode 7 matrix C (signed 8.8; write low,high).
`$211E`	`REG_M7D`	W	Mode 7 matrix D (signed 8.8; write low,high).
`$211F`	`REG_M7X`	W	Mode 7 center X (signed 13-bit; write twice).
`$2120`	`REG_M7Y`	W	Mode 7 center Y (signed 13-bit; write twice).
`$2121`	`REG_CGADD`	W	CGRAM (palette) entry address (0..255).
`$2122`	`REG_CGDATA`	W	CGRAM data port — BGR555, write low byte then high byte.
`$2123`	`REG_W12SEL`	W	Window mask settings for BG1/BG2 (2 bits per window/layer).
`$2124`	`REG_W34SEL`	W	Window mask settings for BG3/BG4.
`$2125`	`REG_WOBJSEL`	W	Window mask settings for OBJ + color-math.
`$2126`	`REG_WH0`	W	Window 1 left position (pixel 0..255).
`$2127`	`REG_WH1`	W	Window 1 right position.
`$2128`	`REG_WH2`	W	Window 2 left position.
`$2129`	`REG_WH3`	W	Window 2 right position.
`$212A`	`REG_WBGLOG`	W	Window-1/2 combine logic per BG (00 OR, 01 AND, 10 XOR, 11 XNOR).
`$212B`	`REG_WOBJLOG`	W	Window combine logic for OBJ + color-math.
`$212C`	`REG_TM`	W	Main-screen layer enable.
`$212D`	`REG_TS`	W	Sub-screen layer enable (bits as TM).
`$212E`	`REG_TMW`	W	Main-screen window-mask enable per layer (bits as TM).
`$212F`	`REG_TSW`	W	Sub-screen window-mask enable per layer (bits as TM).
`$2130`	`REG_CGWSEL`	W	Color-math control window + source.
`$2131`	`REG_CGADSUB`	W	Color-math operation + per-layer enable.

Addr	Name	R/W	Purpose
`$2132`	`REG_COLDATA`	W	Fixed color for color math (write once per channel).
`$2133`	`REG_SETINI`	W	Display mode / interlace settings.
`$2134`	`REG_MPYL`	R	Mode 7 / hw multiply result, low byte (M7A signed16 x M7B signed8 = signed24).
`$2135`	`REG_MPYM`	R	multiply result, middle byte.
`$2136`	`REG_MPYH`	R	multiply result, high byte.
`$2137`	`REG_SLHV`	R	software latch: a dummy read latches the H/V counters into OPHCT/OPVCT.
`$2138`	`REG_OAMDATAREAD`	R	OAM data read (auto-increments OAMADD).
`$2139`	`REG_VMDATALREAD`	R	VRAM data read, low byte (increments per VMAIN).
`$213A`	`REG_VMDATAHREAD`	R	VRAM data read, high byte.
`$213B`	`REG_CGDATAREAD`	R	CGRAM data read (low then high).
`$213C`	`REG_OPHCT`	R	horizontal counter latch (read twice: low then high bit 8; 0..339).
`$213D`	`REG_OPVCT`	R	vertical counter latch (read twice; 0..261 NTSC).
`$213E`	`REG_STAT77`	R	PPU1 (5C77) status.
`$213F`	`REG_STAT78`	R	PPU2 (5C78) status.

▶ Bit fields

[16] = 16-bit access alias over the adjacent L/H byte pair.

DMA / HDMA ($4300–$437F, + $420B/$420C)

Eight transfer channels plus the MDMAEN/HDMAEN enable triggers. — `snes_dma.h`, 90 registers.

Addr	Name	R/W	Purpose
`$420B`	`REG_MDMAEN`	W	General-purpose DMA enable + trigger; CPU stalls until done.

Addr	Name	R/W	Purpose
`$420C`	`REG_HDMAEN`	W	HDMA enable (per-scanline); bit n = arm channel n for HDMA.
`$4300`	`REG_DMAP0`	RW	Channel 0 transfer parameters.
`$4301`	`REG_BBAD0`	RW	Channel 0 B-bus destination (low byte of $21xx, e.g. $18 -> VMDATA).
`$4302`	`REG_A1T0L`	RW	Channel 0 A-bus source address, low byte.
`$4303`	`REG_A1T0H`	RW	Channel 0 A-bus source address, high byte.
`$4304`	`REG_A1B0`	RW	Channel 0 A-bus source bank.
`$4305`	`REG_DAS0L`	RW	Channel 0 byte count low (GP-DMA) / HDMA indirect addr low. 0 count = 65536.
`$4306`	`REG_DAS0H`	RW	Channel 0 byte count high / HDMA indirect addr high.
`$4307`	`REG_DASB0`	RW	Channel 0 HDMA indirect address bank (HDMA only).
`$4308`	`REG_A2A0L`	RW	Channel 0 HDMA table current address, low (HDMA only).
`$4309`	`REG_A2A0H`	RW	Channel 0 HDMA table current address, high (HDMA only).
`$430A`	`REG_NTRL0`	RW	Channel 0 HDMA line counter (HDMA only).
`$4310`	`REG_DMAP1`	RW	Channel 1 transfer parameters (see DMAP0).
`$4311`	`REG_BBAD1`	RW	Channel 1 B-bus destination.
`$4312`	`REG_A1T1L`	RW	Channel 1 A-bus source low.
`$4313`	`REG_A1T1H`	RW	Channel 1 A-bus source high.
`$4314`	`REG_A1B1`	RW	Channel 1 A-bus source bank.
`$4315`	`REG_DAS1L`	RW	Channel 1 byte count low / HDMA indirect low.
`$4316`	`REG_DAS1H`	RW	Channel 1 byte count high / HDMA indirect high.
`$4317`	`REG_DASB1`	RW	Channel 1 HDMA indirect bank.
`$4318`	`REG_A2A1L`	RW	Channel 1 HDMA table address low.

Addr	Name	R/W	Purpose
`$4319`	`REG_A2A1H`	RW	Channel 1 HDMA table address high.
`$431A`	`REG_NTRL1`	RW	Channel 1 HDMA line counter.
`$4320`	`REG_DMAP2`	RW	Channel 2 transfer parameters (see DMAP0).
`$4321`	`REG_BBAD2`	RW	Channel 2 B-bus destination.
`$4322`	`REG_A1T2L`	RW	Channel 2 A-bus source low.
`$4323`	`REG_A1T2H`	RW	Channel 2 A-bus source high.
`$4324`	`REG_A1B2`	RW	Channel 2 A-bus source bank.
`$4325`	`REG_DAS2L`	RW	Channel 2 byte count low / HDMA indirect low.
`$4326`	`REG_DAS2H`	RW	Channel 2 byte count high / HDMA indirect high.
`$4327`	`REG_DASB2`	RW	Channel 2 HDMA indirect bank.
`$4328`	`REG_A2A2L`	RW	Channel 2 HDMA table address low.
`$4329`	`REG_A2A2H`	RW	Channel 2 HDMA table address high.
`$432A`	`REG_NTRL2`	RW	Channel 2 HDMA line counter.
`$4330`	`REG_DMAP3`	RW	Channel 3 transfer parameters (see DMAP0).
`$4331`	`REG_BBAD3`	RW	Channel 3 B-bus destination.
`$4332`	`REG_A1T3L`	RW	Channel 3 A-bus source low.
`$4333`	`REG_A1T3H`	RW	Channel 3 A-bus source high.
`$4334`	`REG_A1B3`	RW	Channel 3 A-bus source bank.
`$4335`	`REG_DAS3L`	RW	Channel 3 byte count low / HDMA indirect low.
`$4336`	`REG_DAS3H`	RW	Channel 3 byte count high / HDMA indirect high.
`$4337`	`REG_DASB3`	RW	Channel 3 HDMA indirect bank.
`$4338`	`REG_A2A3L`	RW	Channel 3 HDMA table address low.

Addr	Name	R/W	Purpose
`$4339`	`REG_A2A3H`	RW	Channel 3 HDMA table address high.
`$433A`	`REG_NTRL3`	RW	Channel 3 HDMA line counter.
`$4340`	`REG_DMAP4`	RW	Channel 4 transfer parameters (see DMAP0).
`$4341`	`REG_BBAD4`	RW	Channel 4 B-bus destination.
`$4342`	`REG_A1T4L`	RW	Channel 4 A-bus source low.
`$4343`	`REG_A1T4H`	RW	Channel 4 A-bus source high.
`$4344`	`REG_A1B4`	RW	Channel 4 A-bus source bank.
`$4345`	`REG_DAS4L`	RW	Channel 4 byte count low / HDMA indirect low.
`$4346`	`REG_DAS4H`	RW	Channel 4 byte count high / HDMA indirect high.
`$4347`	`REG_DASB4`	RW	Channel 4 HDMA indirect bank.
`$4348`	`REG_A2A4L`	RW	Channel 4 HDMA table address low.
`$4349`	`REG_A2A4H`	RW	Channel 4 HDMA table address high.
`$434A`	`REG_NTRL4`	RW	Channel 4 HDMA line counter.
`$4350`	`REG_DMAP5`	RW	Channel 5 transfer parameters (see DMAP0).
`$4351`	`REG_BBAD5`	RW	Channel 5 B-bus destination.
`$4352`	`REG_A1T5L`	RW	Channel 5 A-bus source low.
`$4353`	`REG_A1T5H`	RW	Channel 5 A-bus source high.
`$4354`	`REG_A1B5`	RW	Channel 5 A-bus source bank.
`$4355`	`REG_DAS5L`	RW	Channel 5 byte count low / HDMA indirect low.
`$4356`	`REG_DAS5H`	RW	Channel 5 byte count high / HDMA indirect high.
`$4357`	`REG_DASB5`	RW	Channel 5 HDMA indirect bank.
`$4358`	`REG_A2A5L`	RW	Channel 5 HDMA table address low.

Addr	Name	R/W	Purpose
`$4359`	`REG_A2A5H`	RW	Channel 5 HDMA table address high.
`$435A`	`REG_NTRL5`	RW	Channel 5 HDMA line counter.
`$4360`	`REG_DMAP6`	RW	Channel 6 transfer parameters (see DMAP0).
`$4361`	`REG_BBAD6`	RW	Channel 6 B-bus destination.
`$4362`	`REG_A1T6L`	RW	Channel 6 A-bus source low.
`$4363`	`REG_A1T6H`	RW	Channel 6 A-bus source high.
`$4364`	`REG_A1B6`	RW	Channel 6 A-bus source bank.
`$4365`	`REG_DAS6L`	RW	Channel 6 byte count low / HDMA indirect low.
`$4366`	`REG_DAS6H`	RW	Channel 6 byte count high / HDMA indirect high.
`$4367`	`REG_DASB6`	RW	Channel 6 HDMA indirect bank.
`$4368`	`REG_A2A6L`	RW	Channel 6 HDMA table address low.
`$4369`	`REG_A2A6H`	RW	Channel 6 HDMA table address high.
`$436A`	`REG_NTRL6`	RW	Channel 6 HDMA line counter.
`$4370`	`REG_DMAP7`	RW	Channel 7 transfer parameters (see DMAP0).
`$4371`	`REG_BBAD7`	RW	Channel 7 B-bus destination.
`$4372`	`REG_A1T7L`	RW	Channel 7 A-bus source low.
`$4373`	`REG_A1T7H`	RW	Channel 7 A-bus source high.
`$4374`	`REG_A1B7`	RW	Channel 7 A-bus source bank.
`$4375`	`REG_DAS7L`	RW	Channel 7 byte count low / HDMA indirect low.
`$4376`	`REG_DAS7H`	RW	Channel 7 byte count high / HDMA indirect high.
`$4377`	`REG_DASB7`	RW	Channel 7 HDMA indirect bank.
`$4378`	`REG_A2A7L`	RW	Channel 7 HDMA table address low.

Addr	Name	R/W	Purpose
`$4379`	`REG_A2A7H`	RW	Channel 7 HDMA table address high.
`$437A`	`REG_NTRL7`	RW	Channel 7 HDMA line counter.

▶ Bit fields

CPU I/O — Ricoh 5A22 ($4200–$4217)

Interrupt control, the hardware multiply/divide unit, the H/V IRQ timers, and FastROM select. — `snes_cpu.h` , 25 registers.

Addr	Name	R/W	Purpose
`$4200`	`REG_NMITIMEN`	W	Interrupt + auto-joypad enable.
`$4201`	`REG_WRIO`	W	Programmable I/O port (output). Bit 7 drives the controller-port
`$4202`	`REG_WRMPYA`	W	Unsigned multiply: factor A (8-bit).
`$4203`	`REG_WRMPYB`	W	Unsigned multiply: factor B (8-bit); writing starts 8x8->16, result in RDMPY after ~8 cycles.
`$4204`	`REG_WRDIVL`	W	Unsigned divide: dividend low (16-bit dividend).
`$4205`	`REG_WRDIVH`	W	Unsigned divide: dividend high.
`$4206`	`REG_WRDIVB`	W	Unsigned divide: divisor (8-bit); writing starts 16/8, quotient in RDDIV / remainder in RDMPY after ~16 cycles.
`$4204`	`REG_WRDIV` 16	W	16-bit access alias for the divide dividend (WRDIVL/H).
`$4207`	`REG_HTIMEL`	W	H-count IRQ position, low (0..339).
`$4208`	`REG_HTIMEH`	W	H-count IRQ position, high (bit 0).
`$4209`	`REG_VTIMEL`	W	V-count IRQ position, low (0..261 NTSC).
`$420A`	`REG_VTIMEH`	W	V-count IRQ position, high (bit 0).
`$4207`	`REG_HTIME` 16	W	16-bit access alias for the H IRQ position (HTIMEL/H).

Addr	Name	R/W	Purpose
`$4209`	`REG_VTIME` 16	W	16-bit access alias for the V IRQ position (VTIMEL/H).
`$420D`	`REG_MEMSEL`	W	ROM access speed.
`$4210`	`REG_RDNMI`	R	NMI flag + CPU version.
`$4211`	`REG_TIMEUP`	R	IRQ flag.
`$4212`	`REG_HVBJOY`	R	Blanking + auto-joypad status.
`$4213`	`REG_RDIO`	R	Programmable I/O port (input).
`$4214`	`REG_RDDIVL`	R	Divide quotient, low.
`$4215`	`REG_RDDIVH`	R	Divide quotient, high.
`$4216`	`REG_RDMPYL`	R	Multiply product low (or divide remainder low).
`$4217`	`REG_RDMPYH`	R	Multiply product high (or divide remainder high).
`$4214`	`REG_RDDIV` 16	R	16-bit access alias for the divide quotient (RDDIVL/H).
`$4216`	`REG_RDMPY` 16	R	16-bit access alias for the multiply product / remainder (RDMPYL/H).

▶ Bit fields

16 = 16-bit access alias over the adjacent L/H byte pair.

Controllers / Joypad ($4016–$4017, $4218–$421F)

Serial controller access and the automatic-read pad latches. — `snes_joypad.h`, 14 registers.

Addr	Name	R/W	Purpose
`$4016`	`REG_JOYSER0`	RW	Write: strobe/latch both ports. Read: controller 1 serial bit.
`$4017`	`REG_JOYSER1`	R	Controller 2 serial bit (read).
`$4218`	`REG_JOY1L`	R	Controller 1, low byte.
`$4219`	`REG_JOY1H`	R	Controller 1, high byte.

Addr	Name	R/W	Purpose
`$421A`	`REG_JOY2L`	R	Controller 2, low byte (layout as JOY1L).
`$421B`	`REG_JOY2H`	R	Controller 2, high byte (layout as JOY1H).
`$421C`	`REG_JOY3L`	R	Controller 3 (multitap), low byte.
`$421D`	`REG_JOY3H`	R	Controller 3 (multitap), high byte.
`$421E`	`REG_JOY4L`	R	Controller 4 (multitap), low byte.
`$421F`	`REG_JOY4H`	R	Controller 4 (multitap), high byte.
`$4218`	`REG_JOY1` [16]	R	Controller 1, full 16-bit button word.
`$421A`	`REG_JOY2` [16]	R	Controller 2, 16-bit button word (layout as JOY1).
`$421C`	`REG_JOY3` [16]	R	Controller 3, 16-bit button word.
`$421E`	`REG_JOY4` [16]	R	Controller 4, 16-bit button word.

▶ Bit fields

[16] = 16-bit access alias over the adjacent L/H byte pair.

APU ports — audio mailbox ($2140–$2143)

The four ports to the (separate) SPC700/S-DSP audio processor. — `snes_apu.h`, 4 registers.

Addr	Name	R/W	Purpose
`$2140`	`REG_APUIO0`	RW	APU I/O port 0 (CPU<->SPC700 mailbox; also the command port).
`$2141`	`REG_APUIO1`	RW	APU I/O port 1.
`$2142`	`REG_APUIO2`	RW	APU I/O port 2.
`$2143`	`REG_APUIO3`	RW	APU I/O port 3.

WRAM access port ($2180–$2183)

A CPU window into the full 128 KB of work RAM. — `snes_wram.h`, 4 registers.

Addr	Name	R/W	Purpose
`$2180`	`REG_WMDATA`	RW	WRAM data port; each access auto-increments WMADD.
`$2181`	`REG_WMADDL`	W	WRAM address, low byte.
`$2182`	`REG_WMADDM`	W	WRAM address, middle byte.
`$2183`	`REG_WMADDH`	W	WRAM address, high byte.

▶ Bit fields

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203 registers across 6 subsystems.